

SHATAKSHI AGARWAL

Navi Mumbai, India

MOBILE: 8291463080 | shatakshiagarwal2002@gmail.com | linkedin.com/in/shatakshi-agarwal | github.com/Shatakshitechie

SUMMARY

AI/ML Engineer with an MTech in Artificial Intelligence (DIAT Pune, 8.10 CGPA) and hands-on experience building research-grade AI systems in deep learning, computer vision, and NLP. Developed PACE-Net, a physics-adaptive framework for underwater image enhancement, with a research paper submitted to Signal Processing: Image Communication. GATE DA 2026 qualified (AIR 2024). Proficient in Python, PyTorch, TensorFlow, OpenCV, and Scikit-learn.

EDUCATION

Defence Institute of Advanced Technology (DU)	Jul 2024 - Jun 2026
M.Tech - Computer Science & Engineering (Artificial Intelligence) CGPA: 8.10	Pune, India
Banasthali Vidyapith University	Jul 2020 - May 2024
B.Tech - Computer Engineering CGPA: 8.59	Rajasthan, India

TECHNICAL SKILLS

Languages: Python, Java, C, C++
AI / ML: Machine Learning, Deep Learning, Neural Networks, CNNs, GANs, NLP, Computer Vision, Image Processing, Model Training, Data Preprocessing, Feature Engineering, Pursuing GenAI
Tools & Frameworks: PyTorch, TensorFlow, OpenCV, Scikit-learn, NLTK, Streamlit, Git
Database & Data Handling: SQL Server

EXPERIENCE

C-Dac Pune	July 2023 - Dec 2023
Internship	Pune, India
- Built an AI pipeline for biomedical text processing across multiple document categories, implementing preprocessing, feature engineering and language identification. - Developed a text classification workflow handling biomedical text categorization and language identification using Python and NLP techniques.	
Defence Institute of Advanced Technology	May 2025 - April 2026
Internship	Pune, India
- Developed PACE-Net, a physics-adaptive deep learning framework for underwater image enhancement under a MoES-funded research project. - Conducted comparative analysis of 20+ methods across 4 benchmark datasets, achieving 22+ dB PSNR and 0.87 SSIM.	

PROJECTS

Comparative Analysis and Development of Physics-Adaptive Deep Learning Framework for Underwater Image Enhancement	Machine Learning, Deep Learning, Computer Vision, Image Processing
- Performed a detailed comparative study on commonly used methods & designed PACE-Net with physics-aware priors via Feature-wise Linear Modulation (FiLM), evaluated across 20+ methods and 4 benchmark datasets. - Achieved 22+ dB PSNR and 0.87 SSIM across multiple benchmarks, outperforming existing methods by up to +5 dB. - Research paper "PACE-Net: Physics-Adaptive Conditioning Network for Underwater Image Enhancement" submitted to Signal Processing: Image Communication.	
AI Based Text processing, classification & language identifier for Biomedical text	NLP, Text Classification, Tokenization, Feature Extraction
- Built a Python-based AI pipeline for biomedical text processing, implementing preprocessing, feature extraction and tokenization for text classification. - Developed the text classification module for biomedical text categorization and language identification.	
Data management system to streamline operations for a consulting firm (for the firm - Reena Agarwal & Association)	Python, SQL, Data Management, Excel Automation
- Built a data management application with dedicated modules for customer records, tax data and return filing. - Implemented automated Excel report generation and data search functionalities, reducing manual effort for the firm.	

ACHIEVEMENTS

- Gate DA 2026 score 486, All India Rank 3024
- Gate CSE 2024 score 494, All India Rank 5783
- Participation in a Solar Lamp Assembly world record event at IIT Bombay, Oct' 2018